Expt. No. 2

Title: To study and verification of output and transfer characteristics of PMOS transistor.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

Experiment No. 2

Aim: To study and verification of Output and Transfer characteristics of PMOS transistor.

Apparatus/Software: Computer with P-spice

Link for virtual lab: http://www.vlsi-iitg.vlabs.ac.in

Theory:

Introduction

The metal–oxide–semiconductor field-effect transistor (MOSFET) is a transistor used for amplifying or switching electronic signals. In MOSFETs, a voltage on the oxide-insulated gate electrode can induce a conducting channel between the two other contacts called source and drain. The channel can be of n-type or p-type, and is accordingly called an nMOSFET or a pMOSFET. Figure 1 shows the schematic diagram of the structure of an nMOS device before and after channel formation.

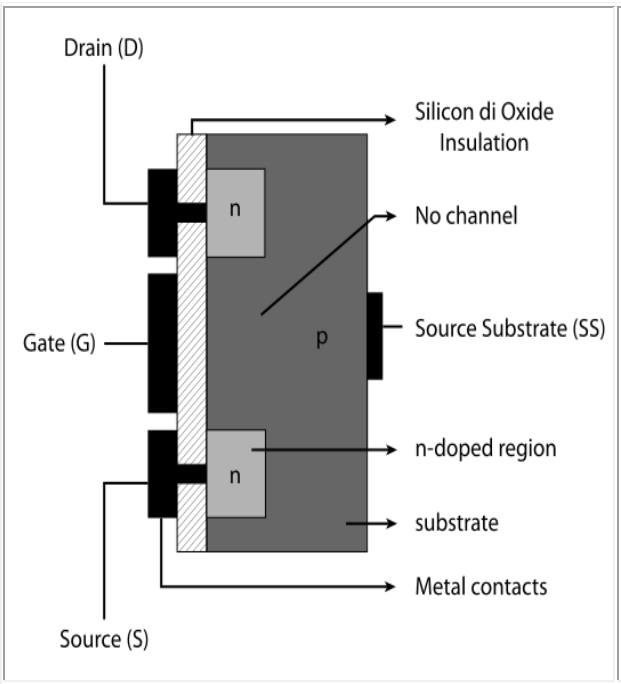


Fig. (1a): nMOSFET before channel formation

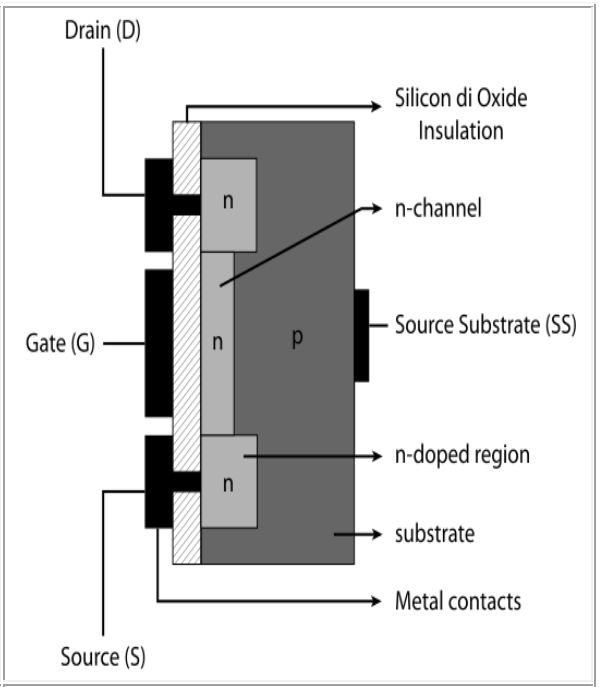


Fig. (1b): nMOSFET structure after channel formation

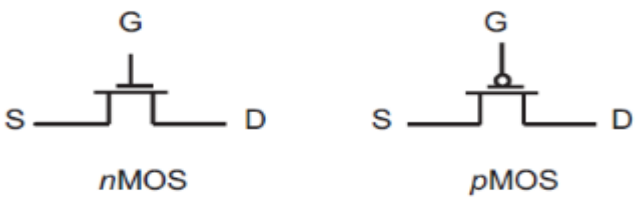
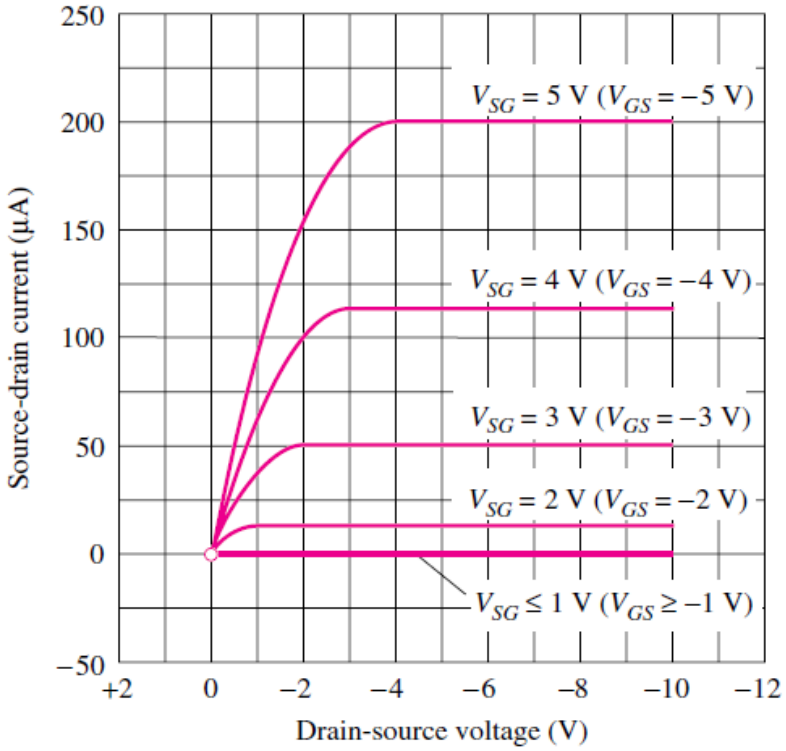


Fig. (2): Circuit symbols for nMOS and pMOS respectively

Output Characteristics

MOSFET output characteristics plot I_D versus V_{DS} for several values of V_{GS}.



P-Channel MOSFET:

The P- Channel MOSFET has a P- Channel region between source and drain. It is a four terminal device such as gate, drain, source, body. The drain and source are heavily doped p+ region and the body or substrate is n-type. The flow of current is positively charged holes. When we apply the negative gate voltage, the electrons present under the oxide layer with are pushed downward into the substrate with a repulsive force. The depletion region populated by the bound positive charges which are associated with the donor atoms. The negative gate voltage also attracts holes from p+ source and drain region into the channel region.

The characteristics of an nMOS transistor can be explained as follows. As the voltage on the top electrode increases further, electrons are attracted to the surface. At a particular voltage level, which we will shortly define as the threshold voltage, the electron density at the surface exceeds the hole density. At this voltage, the surface has inverted from the p-type polarity of the original substrate to an n-type inversion layer, or inversion region, directly underneath the top plate as indicated in Fig. 1(b). This inversion region is an extremely shallow layer, existing as a charge sheet directly below the gate. In the MOS capacitor, the high density of electrons in the inversion layer is supplied by the electron–hole generation process within the depletion layer. The positive charge on the gate is balanced by the combination of negative charge in the inversion layer plus negative ionic acceptor charge in the depletion layer. The voltage at which the surface inversion layer just forms plays an extremely important role in field-effect transistors and is called the threshold voltage V_{tn}. The region of output characteristics where $V_{GS}<vtn</v$ and no

current flows is called the **cutt-off region**. When the channel forms in the nMOS (pMOS) transistor, a positive (negative) drain voltage with respect to the source creates a horizontal electric field moving the electrons (holes) toward the drain forming a positive (negative) drain current coming into the transistor. The positive current convention is used for electron and hole current, but in both cases electrons are the actual charge carriers. If the channel horizontal electric field is of the same order or smaller than the vertical thin oxide field, then the inversion channel remains almost uniform along the device length. This continuous carrier profile from drain to source puts the transistor in a bias state that is equivalently called either the **non-saturated, linear, or ohmic bias state**. The drain and source are effectively short-circuited. This happens when *VGS* > *VDS* + *Vtn* for nMOS transistor and *VGS* < *VDS* +*Vtp* for pMOS transistor. Drain current is linearly related to drain-source voltage over small intervals in the linear bias state.

But if the nMOS drain voltage increases beyond the limit, so that *VGS* < *VDS* + *Vtn*, then the horizontal electric field becomes stronger than the vertical field at the drain end, creating an asymmetry of the channel carrier inversion distribution shown in Figure 4.

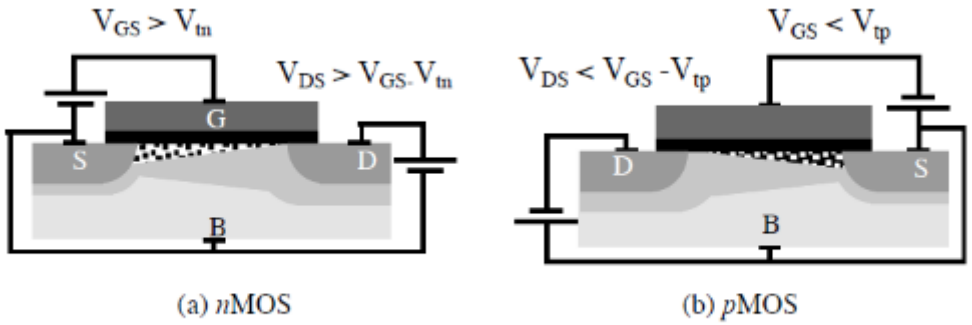


Fig. 4: Channel pinchoff for (a) nMOS and (b) pMOS transistor devices.

If the drain voltage rises while the gate voltage remains the same, then VGD can go below the threshold voltage in the drain region. There can be no carrier inversion at the drain-gate oxide region, so the inverted portion of the channel retracts from the drain, and no longer "touches" this terminal. The pinched-off portion of the channel forms a depletion region with a high electric field. The n-drain and p-bulk form a pn junction. When this happens the inversion channel is said to be "pinched-off" and the device is in the **saturation region**. The characteristics can be loosely modeled by the following equations.

$$I_D = \begin{cases} 0, & \text{Cut-off state} \\ \dfrac{\mu_n \varepsilon_{ox}}{T_{ox}} \dfrac{W}{L}\left[(V_{GS} - V_{tn})V_{DS} - \dfrac{V_{DS}^2}{2}\right], & \text{Ohmic state} \\ \dfrac{\mu_n \varepsilon_{ox}}{2T_{ox}} \dfrac{W}{L}(V_{GS} - V_{tn})^2, & \text{Saturation state} \end{cases}$$

Transfer Characteristics

The transfer characteristic relates drain current (*ID*) response to the input gate-source driving voltage (*VGS*). Since the gate terminal is electrically isolated from the remaining

terminals (drain, source, and bulk), the gate current is essentially zero, so that gate current is not part of device characteristics. The transfer characteristic curve can locate the gate voltage at which the transistor passes current and leaves the OFF-state. This is the device threshold voltage (*Vtn*). Figure 5 shows measured input characteristics for an nMOS and pMOS transistor with a small *0.1V* potential across their drain to source terminals.

The transistors are in their non-saturated bias states. As *VGS* increases for the nMOS transistor in Figure 5a, the threshold voltage is reached where drain current elevates. For *VGS* between *0V* and *0.7V*, *ID* is nearly zero indicating that the equivalent resistance between the drain and source terminals is extremely high. Once *VGS* reaches *0.7V*, the current increases rapidly with *VGS* indicating that the equivalent resistance at the drain decreases with increasing gate-source voltage. Therefore, the threshold voltage of the given nMOS transistor is about $Vtn \approx 0.7V$. The pMOS transistor input characteristic in Figure 5b is analogous to the nMOS transistor except the *ID* and *VGS* polarities are reversed

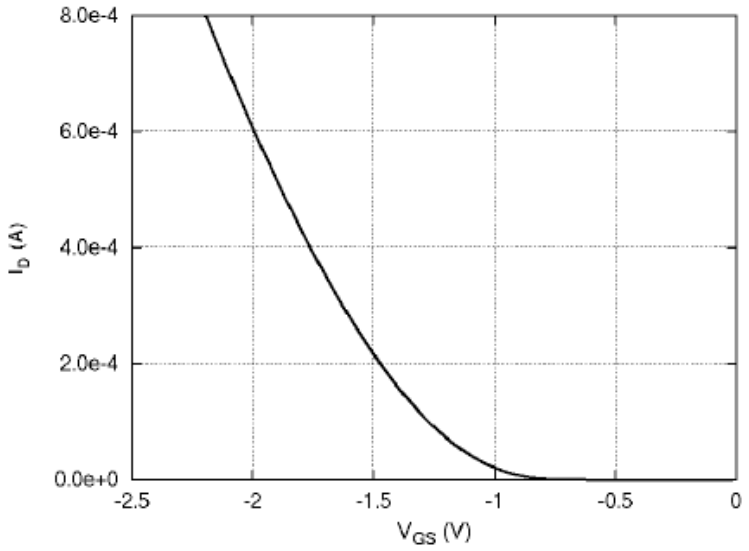
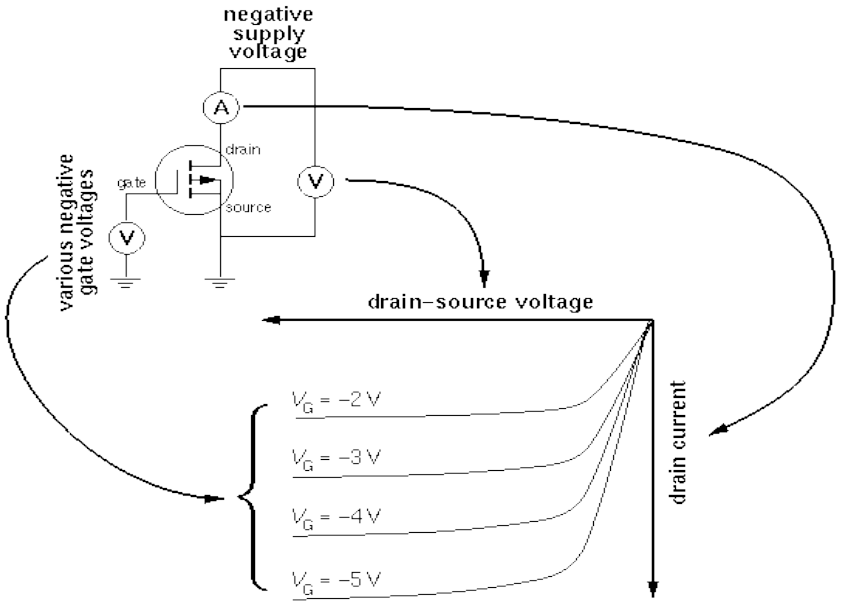


Diagram:



Procedure: Ng-spice

- 1Draw the PMOS circuit.
- Mask the nodes and write the corresponding
- P-spice code in text file. Save it with .cir extension.
- Open it from saved directory.
- Simulate the code and observe the output.
- Record the result obtained.

Procedure: P-spice

PMOS Transfer Characteristics

- Go to File → New File → Text File
- Write the code with the nodes specified as shown
- Specify the model specification of MOS
- Save the file with .cir extension
- Open the file from the directory to make it active
- Run the code
- Add trace to axis of ID(M1)
- Observe the output

PMOS Output Characteristics

- Make changes to the code with the node of VDD and VGS
- Keep the specifications
- Observe the output for trace ID(M1)
- Save the file with .cir extension
- Open the file from the directory to make it active
- Run the code
- Add trace to axis of ID(M1)
- Observe the output.

Program:

PMOS OUTPUT CHARACTERISTICS CODE

```
VGS 1 0 DC 0V

VX 3 2 DC 5v

VDD 3 0 DC 5v

M1 2 1 0 0 PMOS W=1u L=0.35u

.MODEL PMOS PMOS (LEVEL=3

+TOX=7.9E-9 NSUB=1E17 GAMMA=0.4083894

+PHI=0.7 VTO=0.7140674 DELTA=0

+UO=212.23198801 ETA=9.999762E-4 THETA=0.2020774

+KP=6.733755E-5 VMAX=1.181551E5 KAPPA=1.5

+RSH=30.0712458 NFS=1E12 TPG=-1

+XJ=2E-7 LD=5.000001E-13 WD=1.249872E-7

+CGDO=3.09E-10 CGSO=3.09E-10 CGBO=1E-10

+CJ=1.419508E-3 PB=0.8152753 MJ=0.5

+CJSW=4.813504E-10 MJSW=0.5)

.DC VDD 5 -15 -0.01 VGS 5 -5 -1

.PLOT DC I(VX)

.PROBE

.END
```

PMOS TRANSFER CHARACTERISTICS CODE

```
//TRANSFER CHARACTERICTICS OF PMOS

VGS 1 0 DC 0V

VX 3 2 DC 5V

VDD 3 0 DC -15V

M1 2 1 0 0 PMOS W=1u L=0.35u

.MODEL PMOS PMOS (LEVEL=3

+TOX =120E-9 NSUB=1E17 GAMMA=0.40

+PHI=0.7 VTO=0.7 DELTA=0

+UO=212.23198801 ETA=9.999762E-4 THETA=0.2020744

+KP=6.733755E-5 VMAX=1.181551E5 KAPPA=1.5

+RSH=30.0712458 NFS=1E12 TPG=-1

+XJ=2E-7 LD=5.000001E-13 WD=1.249872E-7

+CGDO=3.09E-10 CGSO=3.09E10 CGBO=1E-10

+CJ=1.419508E-3 PB=0.8152753 MJ=0.5

+CJSW=4.813504E-10 MJSW=0.5)

.DC VGS 0 -5 -1

.PLOT DC i(VX)

.PROBE

.END
```

Observations:

The explanation provided above can be summarized in the form of a following table

Kind of MOSFET	Region of Operation		
	Cut-Off	**Ohmic/Linear**	**Saturation**
n-channel	V_{GS}T	$V_{GS} > V_T$ and V_{DS}P	$V_{GS} > V_T$ and $V_{DS} > V_P$

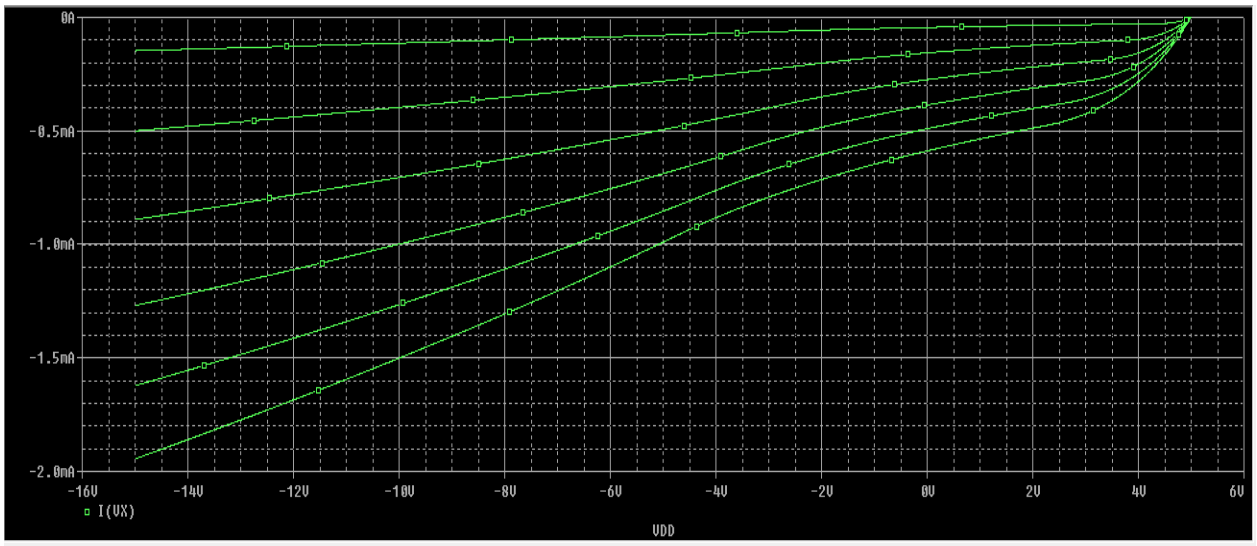
Enhancement-type			
p-channel Enhancement-type	$V_{GS} > -V_T$	$V_{GS}T$ and $V_{DS} > -V_P$	$V_{GS}T$ and $V_{DS}P$
n-channel Depletion-type	$V_{GS}T$	$V_{GS} > -V_T$ and $V_{DS}P$	$V_{GS} > -V_T$ and $V_{DS} > V_P$
p-channel Depletion-type	$V_{GS} > V_T$	$V_{GS}T$ and $V_{DS} > -V_P$	$V_{GS}T$ and $V_{DS}P$

Calculations:

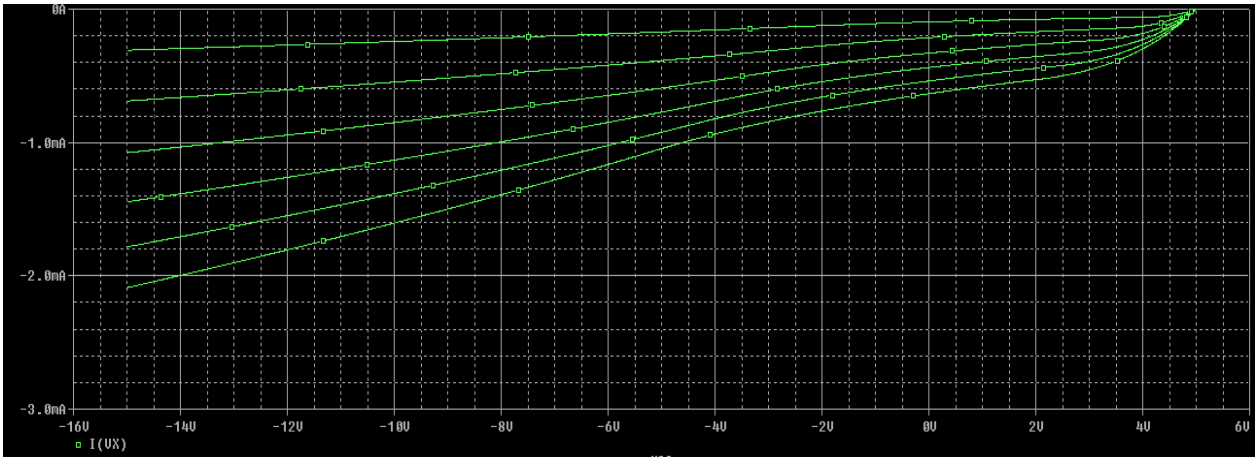
$$I_D = \begin{cases} 0, & \text{Cut-off state} \\ \frac{\mu_n \varepsilon_{ox}}{T_{ox}} \frac{W}{L} \left[(V_{GS} - V_{tn}) V_{DS} - \frac{V_{DS}^2}{2} \right], & \text{Ohmic state} \\ \frac{\mu_n \varepsilon_{ox}}{2T_{ox}} \frac{W}{L} (V_{GS} - V_{tn})^2, & \text{Saturation state} \end{cases}$$

Output (Output Characteristics):

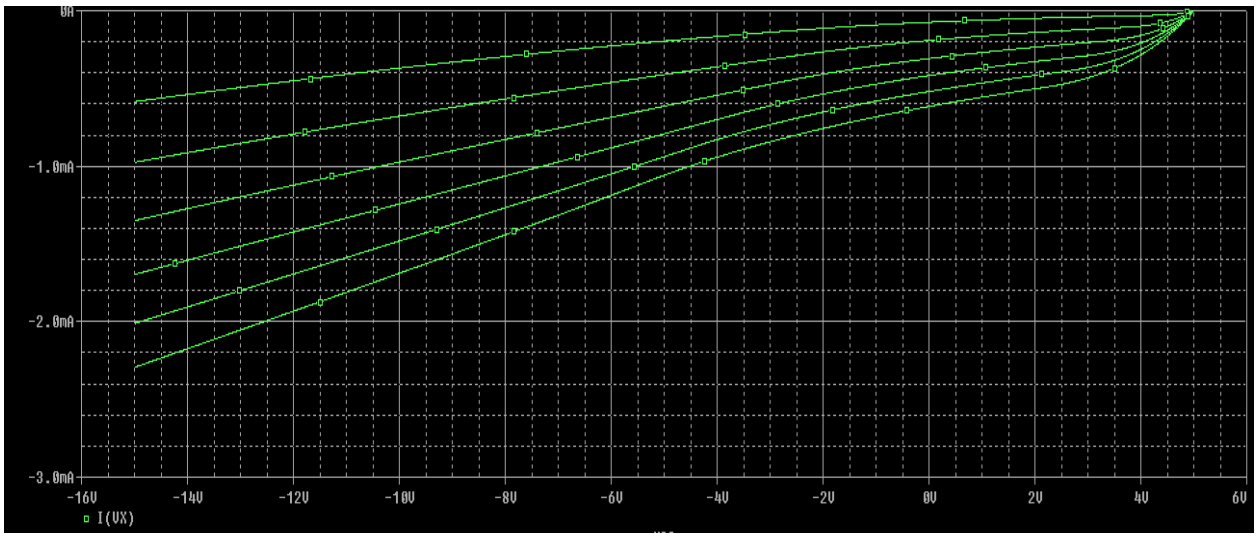
ID=-1.95mA



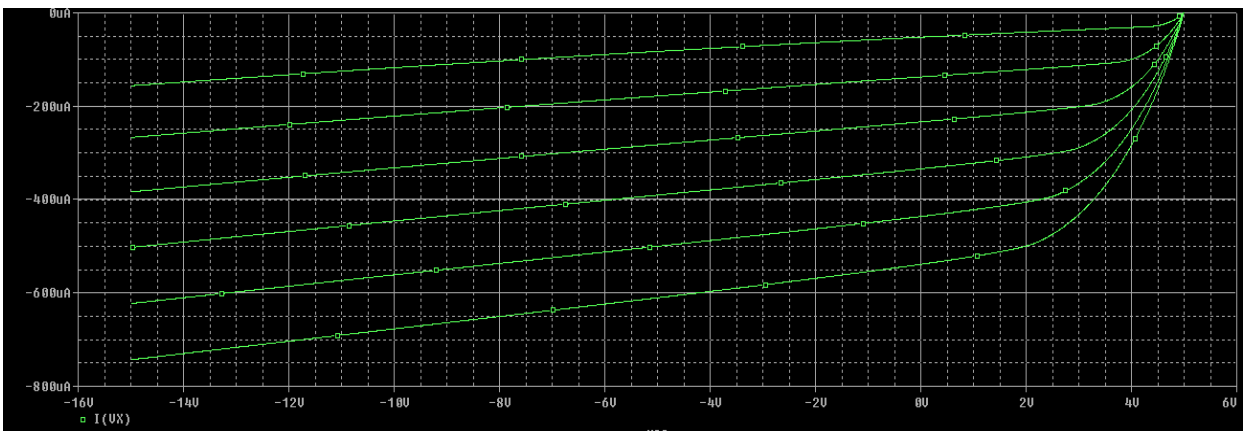
VTO=1.2V ID=-2.1mA



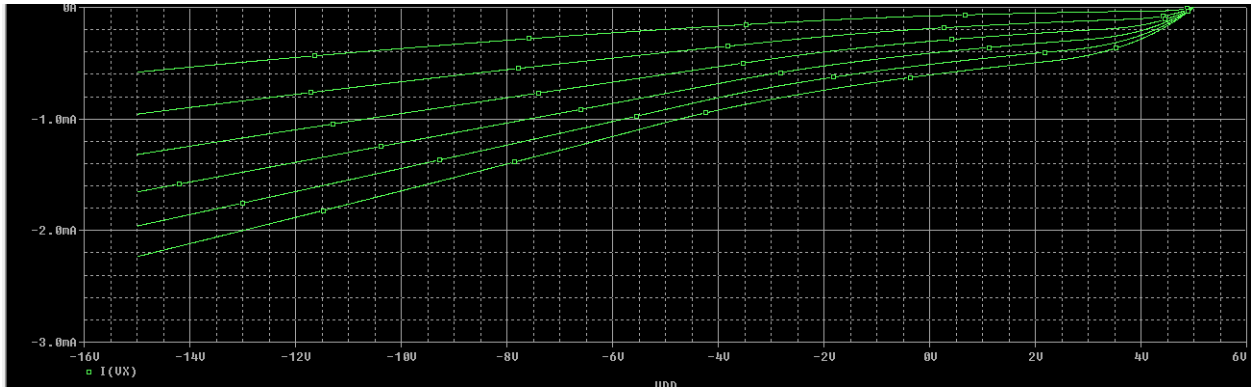
TOX=120E-9 ID=-2.2mA



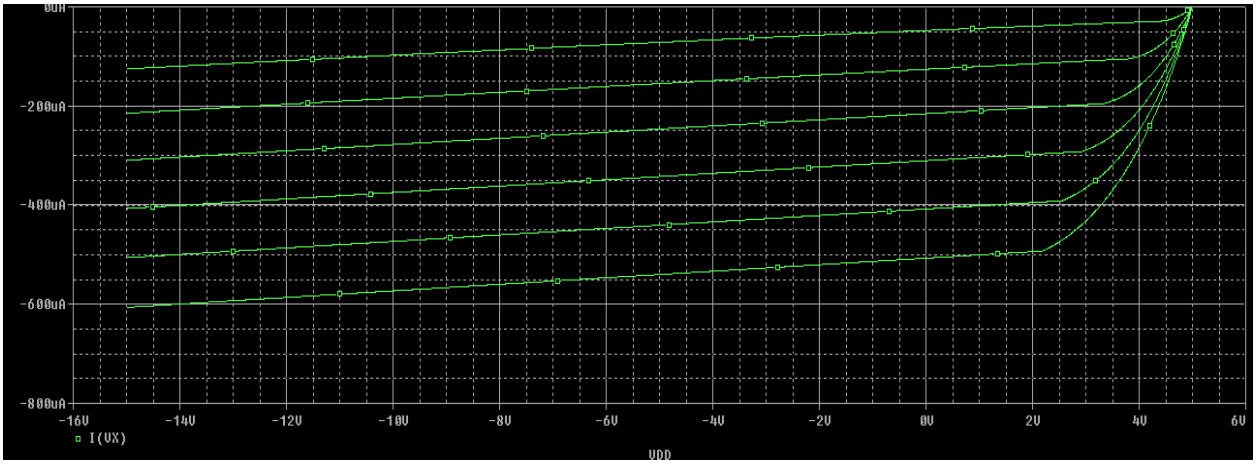
LAMBDA=0.02V-1



GAMMA=0.744 ID=-2.2mA

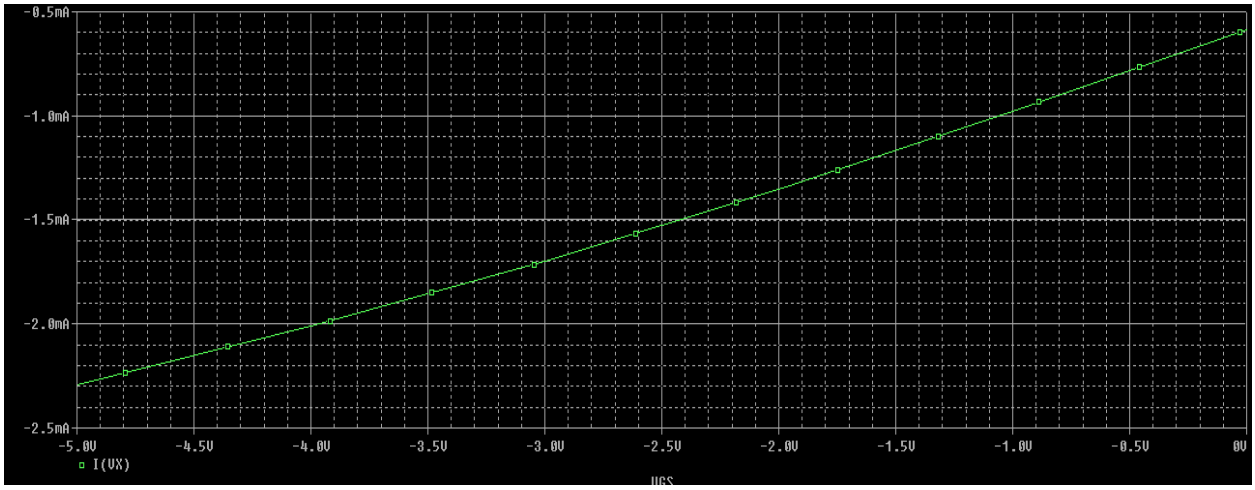


LAMBDA=0V-1 ID=-600mA

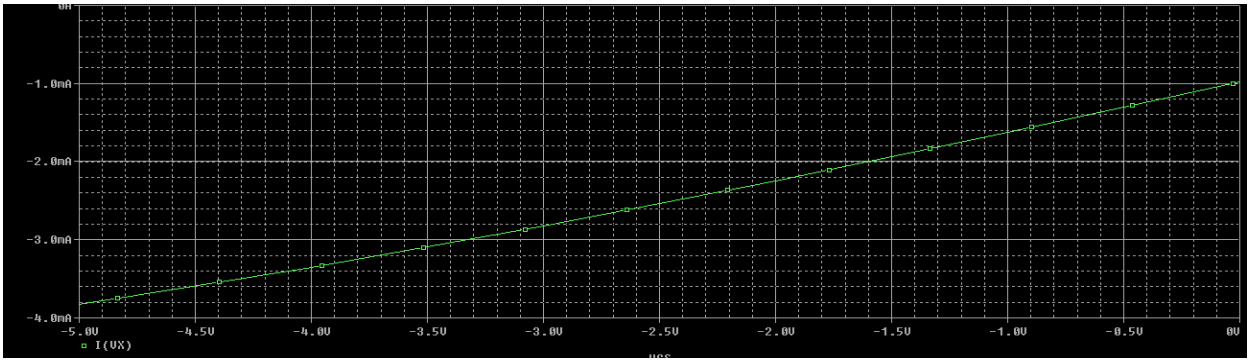


Output (Transfer Characteristics):

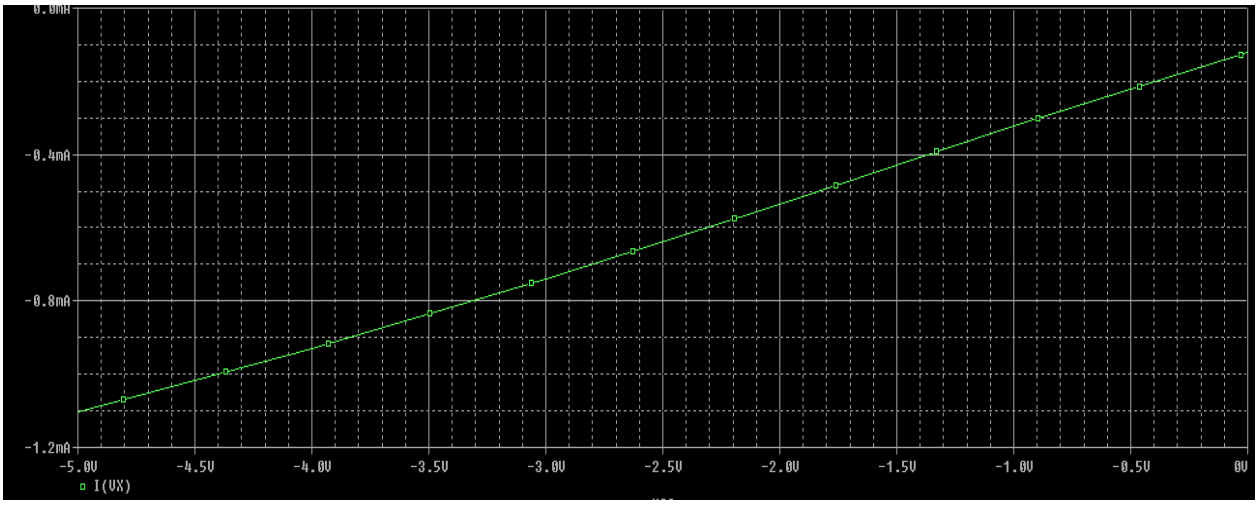
ID=-2.2mA



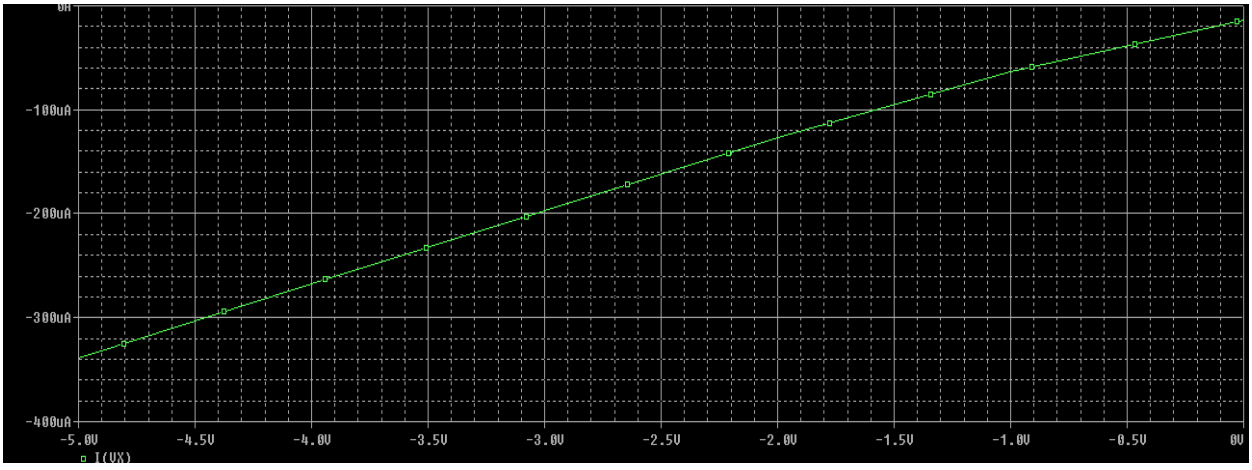
W=1.5u



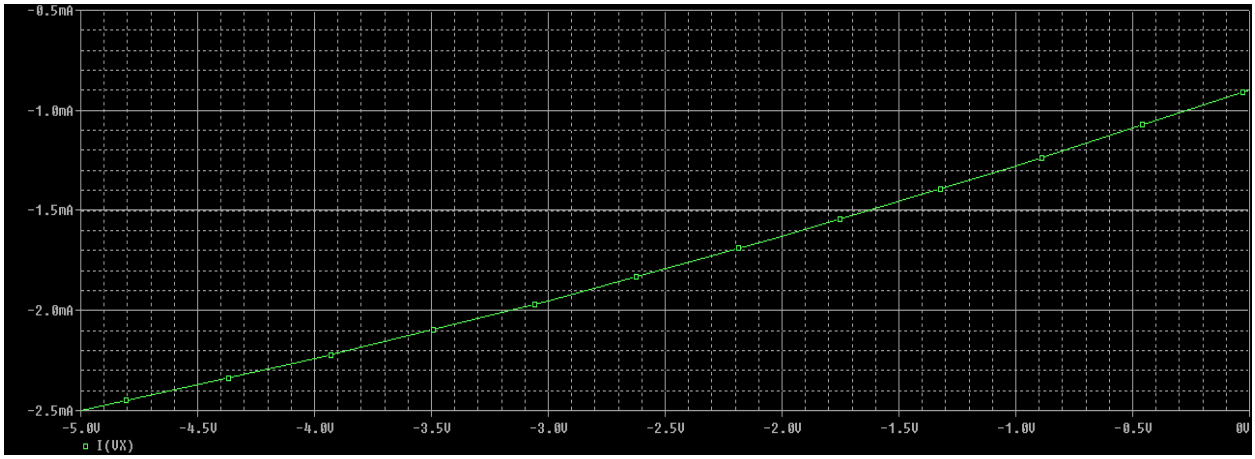
L=0.5u



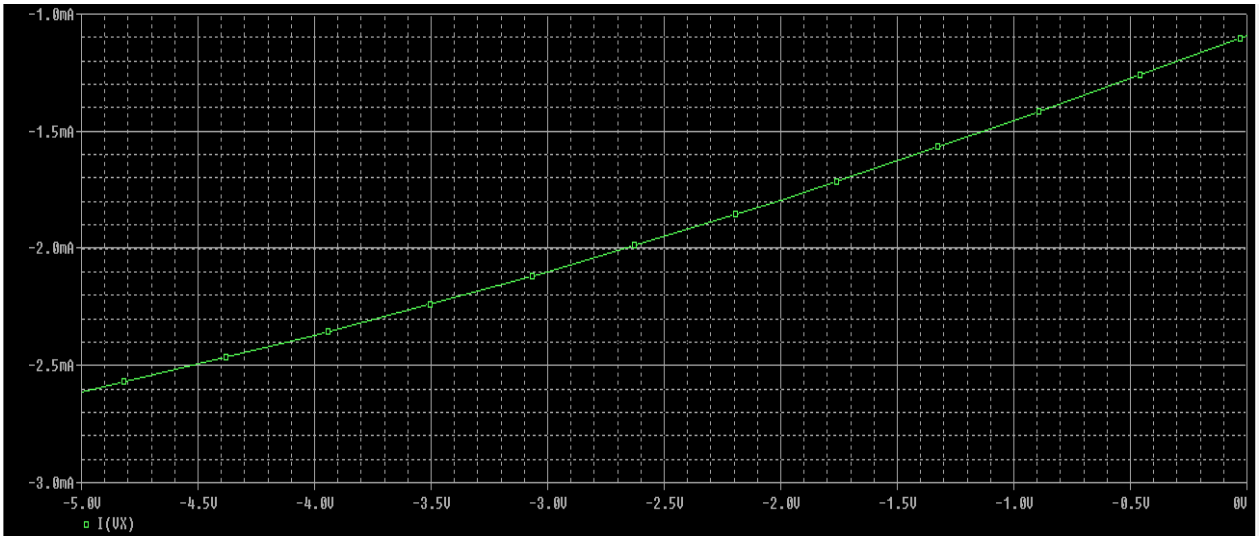
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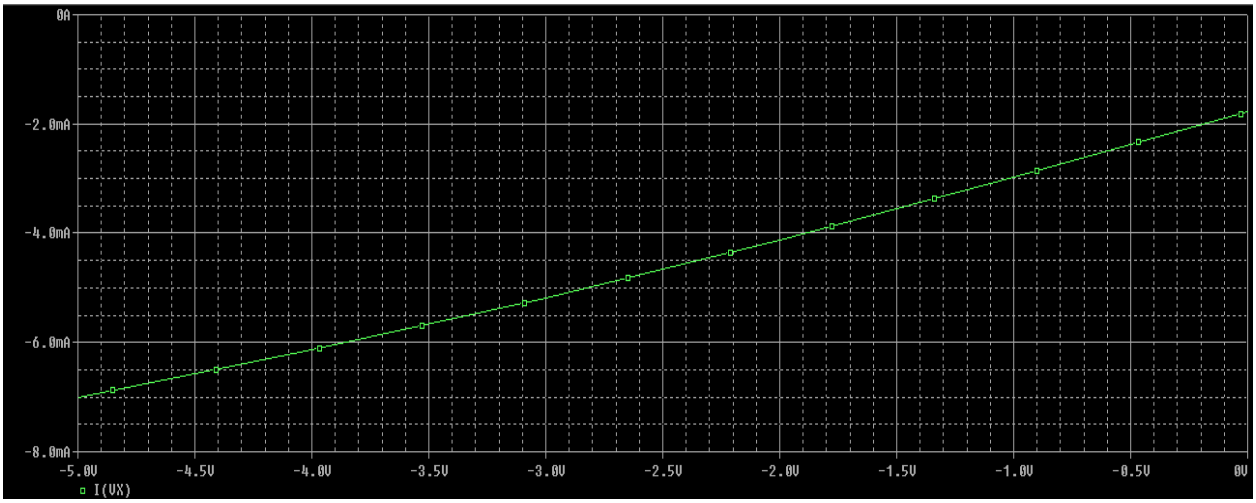
Vto=1.5


-0.5mA
-1.0mA
-1.5mA
-2.0mA
-2.5mA
-5.0V
-4.5V
-4.0V
-3.5V
-3.0V
-2.5V
-2.0V
-1.5V
-1.0V
-0.5V
0V
I(VX)

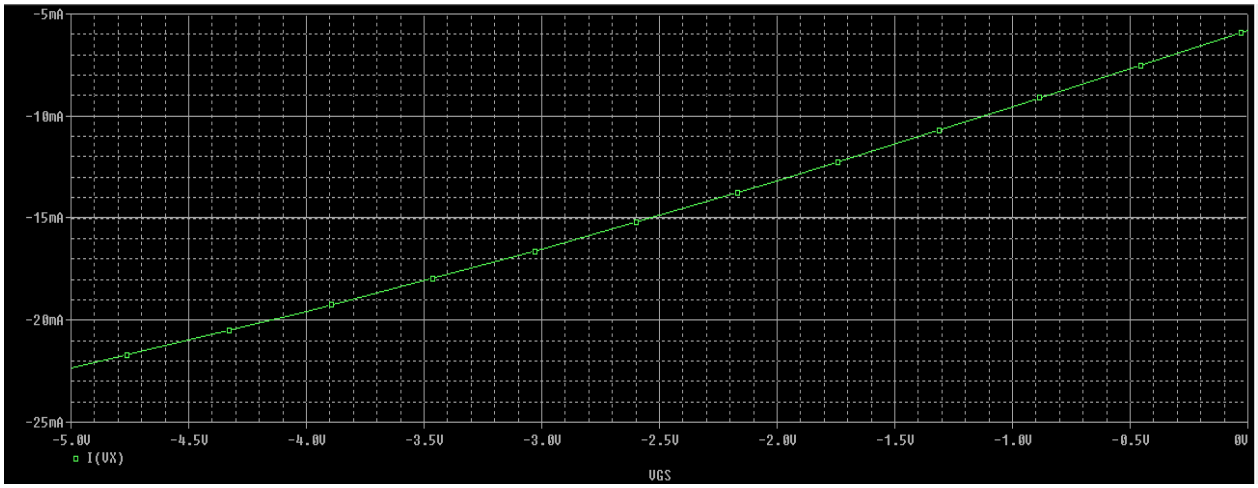
Vto=2


-1.0mA
-1.5mA
-2.0mA
-2.5mA
-3.0mA
-5.0V
-4.5V
-4.0V
-3.5V
-3.0V
-2.5V
-2.0V
-1.5V
-1.0V
-0.5V
0V
I(VX)

Kp=2.055


0A
-2.0mA
-4.0mA
-6.0mA
-8.0mA
-5.0V
-4.5V
-4.0V
-3.5V
-3.0V
-2.5V
-2.0V
-1.5V
-1.0V
-0.5V
0V
I(VX)

Gamma=0.744



Pre Lab Questions:

1. The gate region consists of _____
2. In P channel MOSFET _____ is constant

Post Lab Questions:

. 1. PMOS fabrication process is carried out in _____

Conclusion: Thus, we have studied and experimentally verified the operation of PMOS inverter circuit and recorded the results of output and transfer characteristics

CO's Covered:

……
……